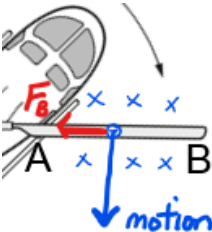


No.	Solution	Remarks
6(a)(i)	$e.m.f = \left -\frac{d\phi}{dt} \right $ $= \frac{\Delta\phi}{t}$ $= \frac{B(\pi r^2)}{T}$ $= \frac{(4.7 \times 10^{-5} \sin 50^\circ)(\pi \times 8^2)}{\left(\frac{1}{200}\right)}$ $= 1.4478$ $\approx 1.4 \text{ V}$	[1] for correct expression/ correct working [1] for correct substitutions [1] for correct answer
6(a)(ii)	 When the rotor blade turns clockwise, electrons in the blade moves across the Earth's magnetic field. By <u>Fleming's left hand rule</u>, a <u>magnetic force towards end A</u> acts on the electrons <u>These electrons move towards A</u> and accumulate there and creates a potential difference between the two ends of the blade as net positive charge accumulate at end B. So <u>end B is at the higher potential</u>	[1] for correct use of Fleming's left hand rule or any correct explanation [1] for correct motion of electrons [1] for correct answer
6(b)(i)	$V_{r.m.s.} = \sqrt{\frac{\frac{1}{2} V_0^2 (0.04)}{0.04}} = \frac{V_0}{\sqrt{2}} = \frac{9}{\sqrt{2}} = 6.3640 \text{ V}$ $P_{\text{mean}} = \frac{V_{r.m.s.}^2}{R} = \frac{6.3640^2}{370} = 0.10946 \approx 0.11 \text{ W}$	[1] for Vrms or any correct working [1] for correct answer
6(b)(ii)	There will only be voltage across the resistor during one half of a cycle (or half-wave rectification). Mean power will be halved.	[1] for any correct explanation [1] for half the power.